

Theorematis arithmetici demonstratio nova

(Gauss, Werke, Band II, articles 5–7)

C. F. Gauss

Re-typeset from scan

5.

THEOREMA. *Sit x quantitas positiva non integra, inter cuius multipla $x, 2x, 3x, \dots$ usque ad nx nullum fiat integer; ponatur $[nx] = h$, unde facile concluditur, etiam inter multipla quantitatis reciprocae $\frac{1}{x}, \frac{2}{x}, \frac{3}{x}, \dots$ usque ad $\frac{h}{x}$ integrum non reperiri. Tum dico fore*

$$\left. \begin{aligned} & [x] + [2x] + [3x] + \dots + [nx] \\ & + \left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + \left[\frac{3}{x}\right] + \dots + \left[\frac{h}{x}\right] \end{aligned} \right\} = nh.$$

Dem. Seriei $[x] + [2x] + [3x] + \dots + [nx]$, quam ponemus $= \Omega$, membra prima usque ad $\left[\frac{1^{\text{tum}}}{x}\right]$ inclus. manifesto omnia erunt $= 0$; sequentia usque ad $\left[\frac{2^{\text{tum}}}{x}\right]$ cuncta $= 1$; sequentia usque ad $\left[\frac{3^{\text{tum}}}{x}\right]$ cuncta $= 2$ et sic porro. Hinc fit

$$\Omega = \left. \begin{aligned} & 0 \times \left[\frac{1}{x}\right] \\ & + 1 \times \left\{ \left[\frac{2}{x}\right] - \left[\frac{1}{x}\right] \right\} \\ & + 2 \times \left\{ \left[\frac{3}{x}\right] - \left[\frac{2}{x}\right] \right\} \\ & + 3 \times \left\{ \left[\frac{4}{x}\right] - \left[\frac{3}{x}\right] \right\} \\ & \text{etc.} \\ & + (h-1) \left\{ \left[\frac{h}{x}\right] - \left[\frac{h-1}{x}\right] \right\} \\ & + h \left\{ n - \left[\frac{h}{x}\right] \right\} \end{aligned} \right\} = hn - \left[\frac{1}{x}\right] - \left[\frac{2}{x}\right] - \left[\frac{3}{x}\right] - \dots - \left[\frac{h}{x}\right].$$

Q. E. D.

6.

THEOREMA. *Designantibus k, p numeros positivos impares inter se primos quoscunque, erit*

$$\left. \begin{aligned} & \left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] + \dots + \left[\frac{\frac{1}{2}(p-1)k}{p}\right] \\ & + \left[\frac{p}{k}\right] + \left[\frac{2p}{k}\right] + \left[\frac{3p}{k}\right] + \dots + \left[\frac{\frac{1}{2}(k-1)p}{k}\right] \end{aligned} \right\} = \frac{1}{4}(k-1)(p-1).$$

Demonstr. Supponendo, quod licet, $k < p$, erit $\frac{\frac{1}{2}(p-1)k}{p}$ minor quam $\frac{1}{2}k$, sed maior quam $\frac{1}{2}(k-1)$, adeoque $\left[\frac{\frac{1}{2}(p-1)k}{p}\right] = \frac{1}{2}(k-1)$. Hinc patet, theorema praesens ex praec. protinus sequi, statuendo illic $\frac{k}{p} = x$, $\frac{1}{2}(p-1) = n$, adeoque $\frac{1}{2}(k-1) = h$.

Ceterum simili modo demonstrari potest, si k fuerit numerus *par* ad p primus, fore

$$\left. \begin{aligned} & \left[\frac{k}{p} \right] + \left[\frac{2k}{p} \right] + \left[\frac{3k}{p} \right] + \cdots + \left[\frac{\frac{1}{2}(p-1)k}{p} \right] \\ & + \left[\frac{p}{k} \right] + \left[\frac{2p}{k} \right] + \left[\frac{3p}{k} \right] + \cdots + \left[\frac{\frac{1}{2}kp}{k} \right] \end{aligned} \right\} = \frac{1}{4}k(p-1).$$

At huic propositioni ad institutum nostrum non necessariae non immoramur.

7.

Iam ex combinatione theorematis praec. cum propos. VIII. art. 4. theorema fundamentale protinus demanat. Nimirum denotantibus k, p numeros primos positivos inaequales quoscunque, et ponendo

$$(k, p) + \left[\frac{k}{p} \right] + \left[\frac{2k}{p} \right] + \left[\frac{3k}{p} \right] + \cdots + \left[\frac{\frac{1}{2}(p-1)k}{p} \right] = L$$

$$(p, k) + \left[\frac{p}{k} \right] + \left[\frac{2p}{k} \right] + \left[\frac{3p}{k} \right] + \cdots + \left[\frac{\frac{1}{2}(k-1)p}{k} \right] = M$$

per VIII. art. 4. patet, L et M semper fieri numeros pares. At per theorema art. 6. erit

$$L + M = (k, p) + (p, k) + \frac{1}{4}(k-1)(p-1).$$

Quoties igitur $\frac{1}{4}(k-1)(p-1)$ par evadit, quod fit, si vel uterque k, p vel saltem alterutur est formae $4n+1$, necessario (k, p) et (p, k) vel ambo pares vel ambo impares esse debent. Quoties autem $\frac{1}{4}(k-1)(p-1)$ impar est, quod evenit, si uterque k, p est formae $4n+3$, necessario alter numerorum $(k, p), (p, k)$ par, alter impar esse debebit. In casu priori itaque relatio ipsius k ad p et relatio ipsius p ad k (quatenus alter alterius residuum vel non-residuum est) identicae erunt, in casu posteriori oppositae. Q. E. D.

Audit notes: The reader (earlier source draft) version of this passage introduced a spurious factor of 2 in the multiples ($2\lambda k$ instead of λk , $2\mu p$ instead of μp), put an undefined z in place of $\frac{1}{4}(k-1)(p-1)$, and replaced denominator p with $\frac{1}{2}p$ and upper limit $\frac{1}{2}(p-1)$ with $\frac{1}{4}(p-1)$. All four errors are corrected here against the scan.